

Information Technology University of the Punjab
FoE Final Examination (Fall 2025)
SE460T B – Cloud Computing with Big Data Applications

Name: _____

Roll No.: _____

Time allowed: 120 minutes

Maximum Marks: 75

Date & Day: 14th January, 2025 (Wednesday)

Instructions

- This exam will assess your CLOs as per OBE. The CLOs are mentioned below.
- This is a **CLOSED BOOK** and **CLOSED NOTES** exam.
- The exam has **TWO parts**.
- **PART I** has **THIRTY-EIGHT (38) MCQs**. Mark your choice from the given options with a tick (✓). Some questions have more than one correct options. This is mentioned in the question statement where applicable.
- **PART II** has **THIRTEEN (13) short questions**. Provide concise answers in the space provided.
- If you are found cheating (use of unfair means, unauthorized material or unauthorized equipment) or helping others cheat, your exam will be cancelled immediately and disciplinary action will be taken.

CLOs

1. Be able to describe challenges with Big Data Analysis, different types of cloud platforms, different reasons for adopting the cloud solution, and the challenges with these different reasons.
 2. Be able to reason about solutions to the common challenges with the cloud solutions.
 3. Be able to implement and configure a big data analysis, including configuring the cloud platform and (if applicable) database.
 4. Feasibility study to migrate existing applications to a cloud environment from both a technical and an economic point of view.
-

PART I – CLO 1	PART II – CLO 2	TOTAL
/ 40	/ 35	/ 75

Instructor: Engr. Muhammad Umair Shoaib
Teaching Assistant: Muhammad Moiz Ahmad

Signature: _____

PART I

1. Which of the following statements that compare data analytics and AI/ML is incorrect? [1]
 - a. AI/ML uses rule based logic whereas data analytics uses mathematical models for predictions
 - b. Data analytics is good for structured data and AI/ML is good for unstructured data
 - c. Data used by data analytics generally have fewer variables compared to the data used by AI/ML
 - d. AI/ML is generally superior to data analytics in making predictions
2. An e-commerce company analyzes last month's sales reports to identify why revenue dropped in a specific region by drilling down into product categories and customer segments. Which type of analytics is being performed? [1]
 - a. Descriptive analytics
 - b. Diagnostic analytics
 - c. Predictive analytics
 - d. Prescriptive analytics
3. A logistics company uses historical delivery data and machine learning models to forecast delivery delays and automatically recommends the optimal rerouting strategy to minimize late shipments. Which type of analytics best describes this scenario? [1]
 - a. Descriptive analytics
 - b. Diagnostic analytics
 - c. Predictive analytics
 - d. Prescriptive analytics
4. Which statement best explains why organizations often place higher value on real-time data compared to older historical data? [1]
 - a. Real-time data is always cheaper to store and easier to manage than historical data
 - b. Real-time data enables immediate, proactive actions, while older data mainly supports reactive analysis and reporting
 - c. Historical data cannot be used for analytics once it becomes old
 - d. Data value depends only on how much data an organization stores, not on processing speed or timing
5. A financial services company is designing a data platform to support real-time fraud detection. Achieving near-perfect accuracy would require significantly higher infrastructure costs and slower processing. Which trade-off best reflects a sound data engineering decision in this scenario? [1]
 - a. Prioritize maximum accuracy, even if detection is delayed and costs increase
 - b. Balance processing speed and acceptable accuracy while controlling infrastructure cost
 - c. Minimize cost by processing data in large daily batches
 - d. Collect and store all data first, then optimize speed and accuracy later
6. Which of the following questions is most likely to be asked by a data engineer when designing a data pipeline? [1]
 - a. Which machine learning model best fits the data?
 - b. How will the analytical results be evaluated and visualized?
 - c. What are the security requirements, data sources, formats, and ingestion mechanisms for this data?
 - d. What insights can the data provide to the business?
7. Which component of HDFS manages metadata such as file names, block locations, and permissions? [1]
 - a. DataNode
 - b. ResourceManager
 - c. NameNode
 - d. NodeManager
8. Which YARN component is responsible for accepting job submissions and launching the first container for an application? [1]
 - a. Scheduler
 - b. Node Manager
 - c. Application Manager
 - d. Application Leader
9. YARN acts as a distributed operating system by managing resources across the entire Hadoop cluster. [1]
 - a. True
 - b. False

10. Which statement best captures the main value proposition of the MapReduce programming model? [1]
- a. It provides low-latency access to transactional data
 - b. It abstracts distributed execution while requiring users to manage fault tolerance
 - c. It allows users to define Map and Reduce logic while the framework manages parallelism and failures
 - d. It is primarily designed for real-time stream processing
11. Which of the following is a valid output of a Map function in a word count application? [1]
- a. (document, total_words)
 - b. (word, 1)
 - c. (word, total_count)
 - d. (file, frequency_table)
12. Which phase of MapReduce is responsible for transferring and grouping mapper output so that all values for the same key reach the same reducer? [1]
- a. Mapper phase
 - b. Partitioning phase
 - c. Shuffle & Sort phase
 - d. Reducer phase
13. Apache Flink supports both batch processing and streaming processing through different APIs. [1]
- a. True
 - b. False
14. Which of the following are higher-level data processing frameworks that automatically generate MapReduce jobs? Select two. [2]
- a. Apache Pig
 - b. Apache Hive
 - c. Apache Spark
 - d. Apache HBase
15. A data lake typically stores highly curated data that serves as the single version of truth. [1]
- a. True
 - b. False
16. Which statement correctly compares schema handling in a data warehouse and a data lake? [1]
- a. Both use schema-on-read
 - b. Data warehouse uses schema-on-read; data lake uses schema-on-write
 - c. Data warehouse uses schema-on-write; data lake uses schema-on-read
 - d. Both require schema to be defined during data ingestion
17. What is the primary purpose of AWS Lake Formation? [1]
- a. To provide low-latency transactional databases
 - b. To convert an S3-based data lake into a secure, governed analytics platform
 - c. To replace Amazon S3 for data storage
 - d. To execute SQL queries on streaming data
18. Which of the following statements about the Leader Node in Amazon Redshift is TRUE? [1]
- a. It stores all the data for the cluster.
 - b. It coordinates queries and creates execution plans.
 - c. It processes queries directly on stored data.
 - d. It has multiple node slices for parallel processing.
19. Which statement best describes Redshift Spectrum? [1]
- a. It copies all S3 data into Redshift before querying.
 - b. It allows querying data directly in S3 without loading it into Redshift.
 - c. It only works with Redshift internal tables.
 - d. It replaces AWS Glue for metadata management.
20. Which characteristic of ELT most strongly favors choosing it over ETL? [1]
- a. Fixed schema requirements
 - b. Need to eliminate PII early
 - c. Frequent changes in analytical requirements
 - d. Limited storage availability

21. Which of the following best describes the primary trade-off of ETL? [1]
- a. Faster ingestion but slower querying
 - b. Slower ingestion but faster querying
 - c. Faster ingestion and faster querying
 - d. Slower ingestion and slower querying
22. A data engineer is creating a stream processing pipeline that needs to reformat incoming data from .csv to .json before delivering it to an S3 bucket. They are looking for the option that requires the least amount of coding. Which service should the data engineer use for this use case? [1]
- a. Use Amazon Kinesis Data Streams.
 - b. Use an Amazon Managed Service for Apache Flink application.
 - c. Use Amazon Data Firehose.
 - d. Use the Amazon Kinesis Client Library (KCL).
23. A traditional retailer with a relatively small online presence recently decided that they need to expand their online presence to increase sales. After a major marketing push and investing in IT infrastructure to support increased online transactions, quarterly results show that online sales results are far below expectations. The business wants to find ways to improve the poor online sales performance. Which action represents an appropriate first step to address the request in a data-driven way? [1]
- a. Develop an ML model that analyzes customer comments to determine where customers are unsatisfied.
 - b. Use A/B testing to provide some customers with a new UI that is designed to simplify the purchasing process. Then, compare sales results from the old and new UIs.
 - c. Use a business intelligence (BI) tool to analyze sales data for the previous two quarters. Then, generate a hypothesis as to why the online sales are not in line with expectations.
 - d. Analyze the total revenue per customer, and create customer segments that receive different types of offers based on their segment.
24. A data engineer has been asked to provide data to support a new sales report. The report will combine sales data for four different products, which have been tracked in four different systems. The resulting data needs to be segmented into regions, and each regional dataset should be available only to sales teams for the region. What should the data engineer do first to meet this request? [1]
- a. Create an S3 bucket with folders for each product.
 - b. Export data from each system, and merge the data into a single file.
 - c. Identify the relationships between fields across each of the sources.
 - d. Replace null values in each source system field with zeroes.
25. Which of the following statements best describes the role of Amazon Athena and AWS Glue in an AWS analytics architecture? [1]
- a. Both Amazon Athena and AWS Glue are used to query data stored in Amazon S3
 - b. AWS Glue is used for querying data, while Amazon Athena transforms the data
 - c. Both Amazon Athena and AWS Glue primarily perform data ingestion from source systems
 - d. Amazon Athena is used for querying data, while AWS Glue is used for ETL and metadata cataloging

26. Suppose the size of our yellow table is 10 GB. We run the following Athena query to create a view. [1]
- ```
CREATE VIEW cctrips AS
SELECT "sum"("fare") "CreditCardFares"
FROM yellow
WHERE ("paytype"='1');
```

Then, we run the following query:

```
Select * from cctrips;
```

How much data would be scanned?

- a. Much less than 10 GB
- b. Equal to 10 GB
- c. Depends on the number of rows in “yellow” where paytype is 1.
- d. 10 GB in first run of this query, and then will reduce after each run

27. Which operation in data lakes is enhanced by using Apache Hudi? [1]
- a. Incremental inserts and updates
  - b. Read-only queries
  - c. Manual schema creation
  - d. Real-time dashboards
28. Apache Hudi is an AWS-managed service. [1]
- a. True
  - b. False
29. In an AWS architecture, where is the Apache Hudi Connector for AWS Glue typically used? [1]
- a. Between Amazon S3 and Amazon Athena, to enable in-place updates and schema evolution
  - b. As a query engine, replacing Athena
  - c. Directly in Amazon Kinesis, to generate streaming data
  - d. As a replacement for AWS Glue Data Catalog
30. Four of the following services have only serverless option. Select them: [2]
- a. AWS Lambda
  - b. Amazon Athena
  - c. Amazon Redshift
  - d. Amazon EMR
  - e. Amazon RDS
  - f. AWS Glue
  - g. Amazon EC2
  - h. Amazon OpenSearch Service
31. You have a large dataset stored in Amazon S3. You need to run queries for both ad-hoc analysis and repeated, complex analytics. Which of the following statements are correct? Select all that apply. [1]
- a. Amazon Athena is better suited for ad-hoc queries directly on S3 without moving data.
  - b. Amazon Redshift is better suited for frequent, complex queries on structured data stored internally.
  - c. Amazon Athena is always faster than Redshift for large-scale, repeated analytics.
  - d. Amazon Redshift requires loading and storing data in its warehouse, but this enables consistent high-performance queries.
  - e. Athena automatically optimizes repeated query performance better than Redshift.
32. What is the primary role of AWS CloudFormation in cloud architectures? [1]
- a. To run SQL queries on AWS data warehouses
  - b. To store large amounts of structured and unstructured data
  - c. To provide a serverless compute environment for ETL jobs
  - d. To provision and manage AWS resources automatically using infrastructure as code
33. Which statement correctly describes the difference between a CloudFormation template and a CloudFormation stack? [1]
- a. A template is a running set of AWS resources; a stack is the blueprint for creating them.
  - b. A template is a JSON or YAML file that defines resources; a stack is the actual set of resources created from that template.
  - c. A stack stores templates in S3; a template executes resources in AWS.
  - d. There is no difference; template and stack are interchangeable terms.
34. Which of the following best describes bucketization in a data lake? [1]
- a. Dividing data into directories based on column values for faster filtering
  - b. Dividing data into a fixed number of files (buckets) using a hash function for better parallelism
  - c. Storing data in different storage classes based on frequency of access
  - d. Compressing data files to reduce storage costs
35. In OpenSearch Service, what is an index? [1]
- a. A single log file stored on S3
  - b. The operating system of the EC2 instance
  - c. A type of Kinesis Data Firehose delivery stream
  - d. A structure that organizes data for fast search and retrieval

36. What is the primary format of data stored in OpenSearch? [1]
- a. CSV
  - b. XML
  - c. Parquet
  - d. JSON
37. What is the default block size in HDFS? [1]
- a. 32 MB
  - b. 64 MB
  - c. 128 MB
  - d. 256 MB
38. Redshift Spectrum requires the data in S3 to be registered in a catalog, such as the AWS Glue Data Catalog. [1]
- a. True
  - b. False

## PART II

39. Arrange the following MapReduce stages in the correct execution order:

- a. Reducer aggregates results and writes output
- b. Shuffle & Sort groups intermediate keys
- c. Mapper produces key-value pairs
- d. Job is divided into parallel tasks

Write the correct sequence as letters.

[2]

d c b a

40. Name one type of analytics workload that is more suitable for a data lake than a data warehouse.

[1]

AI/ML workload with unstructured data

41. Which feature of AWS Glue Studio can be used by users with limited coding experience to author jobs?

[1]

Visual ETL

42. List the six steps involved in data wrangling.

[3]

Discovery, Structuring, Cleaning, Enriching, Validating, Publishing

43. Provide the names of four AWS services that can be used for ingestion, along with the sources of data that each can ingest from.

[4]

AppFlow : SaaS apps

AWS DMS : Relational databases

DataSync: File Systems

AWS Data Exchange: Third-party datasets

44. List the three broad categories of data sources with respect to variety.

[3]

On-premise databases

Public datasets

Events/IOT devices

45. What is meant by data veracity? Give your answer in one sentence. [1]

How accurate and trustworthy the data is.

46. Give one reason why a data engineer might reduce transformations before loading data into a lake. [1]

If downstream users need raw data.

There can be multiple correct answers for this question.

47. What is the role of Amazon Data Firehose in a data pipeline? How does Data Firehose support built-in transformations? [2]

To move data from Kinesis Data Streams to S3.

We can use a Lambda function integrated within Firehose to perform transformations.

48. What are the main stages of a data engineering pipeline? Also, for each stage of the pipeline, provide names of two AWS services that can perform the role of that stage. [8]

Ingestion: AWS Glue, Kinesis Datastreams

Storage: S3, Redshift

Processing: Amazon EMR, Athena, Glue Jobs

Analysis and Visualizatin: Athena, Opensearch

49. Which component of the Hadoop ecosystem is replaced by Spark, and why is Spark generally faster for querying than this component? [2]

MapReduce.

In-memory computation and processing.



50. At what stage of the pipeline do we use Amazon Kinesis Data Streams? What is a shard in a Kinesis Data Stream? What is the default retention period for data in Kinesis Data Streams? [3]

Ingestion

A shard is a unit of capacity that provides a fixed amount of read and write throughput and stores a sequence of data records in order.

24 hours

51. Briefly compare data warehouses and data lakes in terms of data quality, storage cost and query performance. [3]

Warehouse vs Lake

Quality: Highly curated, structured, semi-structured data ----- any data, may be curated or raw.

Storage cost: High ----- low

Query performance: Fastest query results ----- Slower compared to the data warehouse

52. Name two AWS services that can store a data catalog. [1]

Redshift

Glue